北京大學
PEKING UNIVERSITY

第六章 固体理论

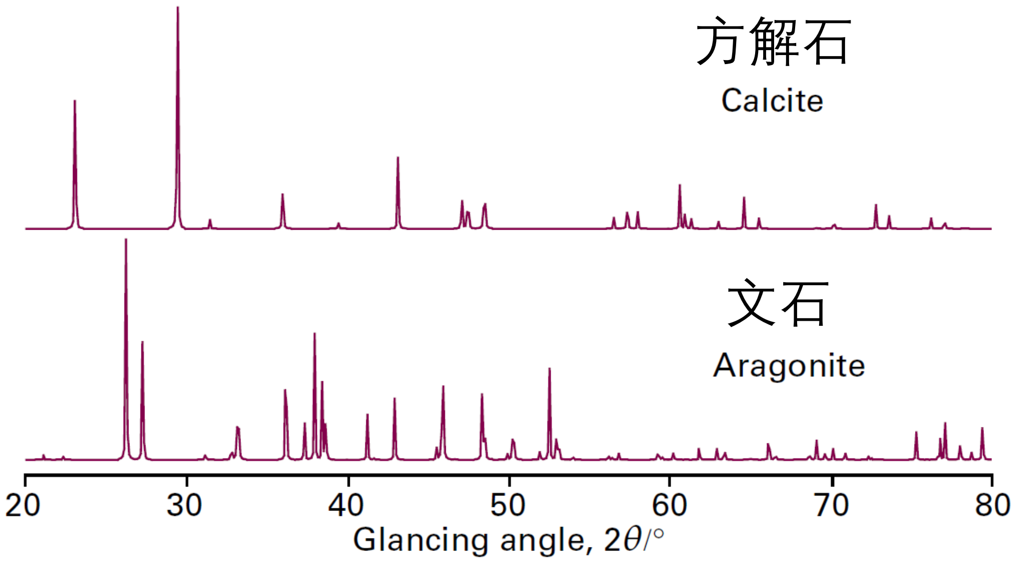
2. 晶体的衍射

X射线晶体学

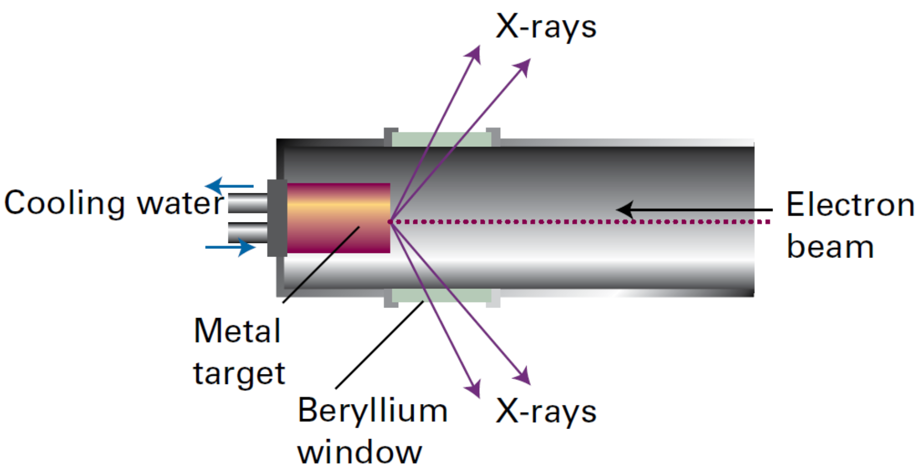
X射线衍射（X-ray diffraction, XRD）

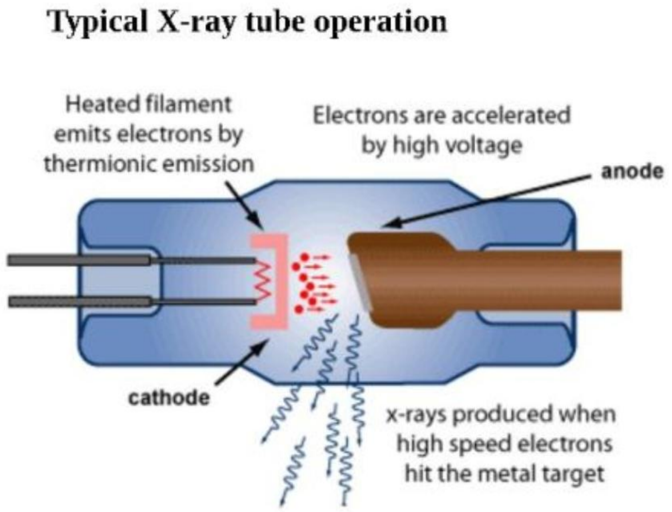
衍射：遇到障碍物之后的干涉

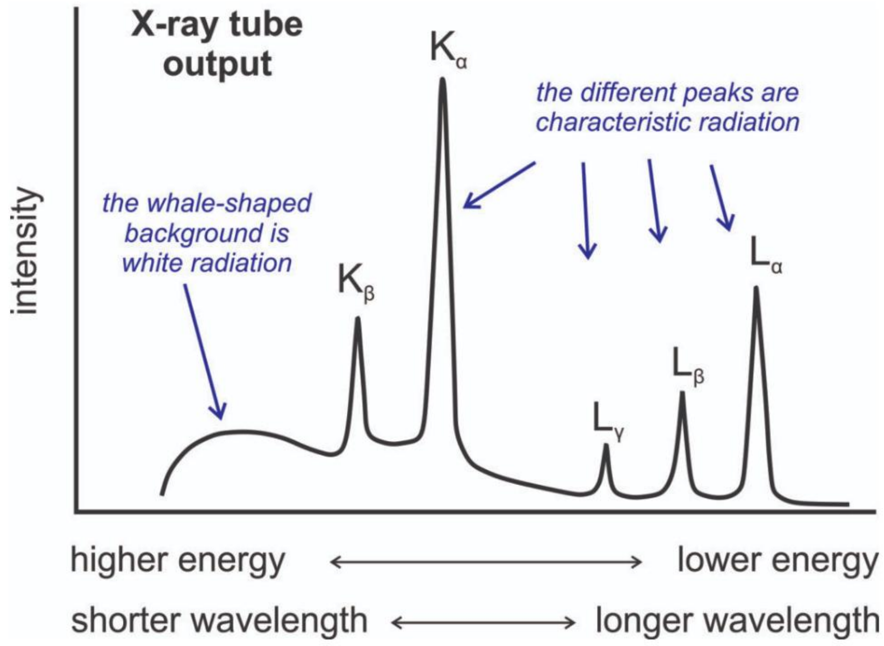
X射线波长~100pm，与晶面间距相当

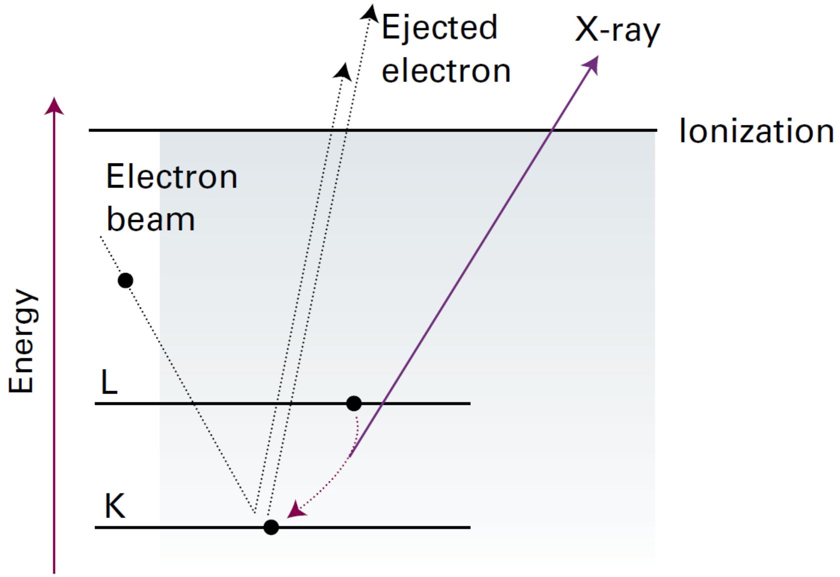


两种$CaCO_3$ X射线粉末衍射图样









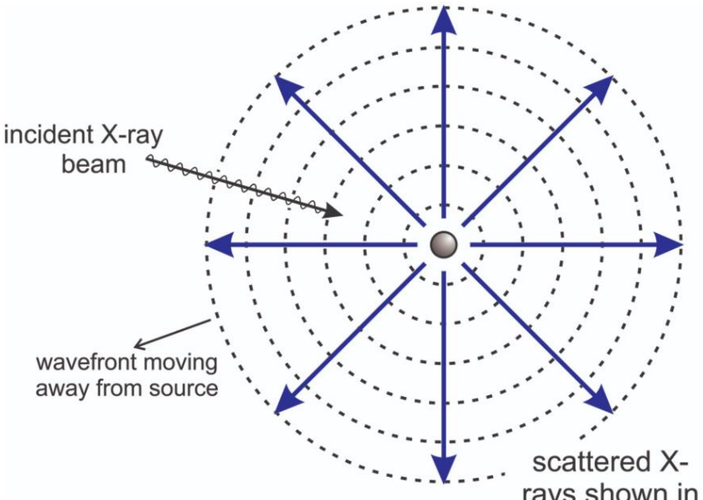
X射线发射谱

最强峰对应K_α射线

常用Cu K_α射线

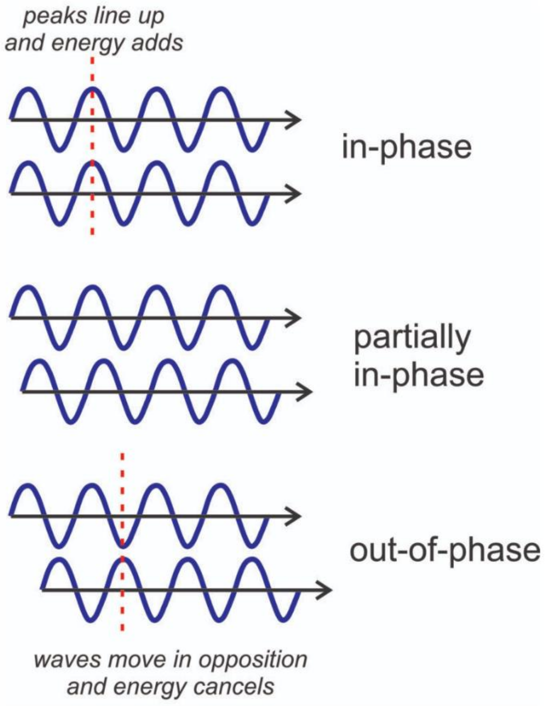
$\lambda = 1.5418Å$

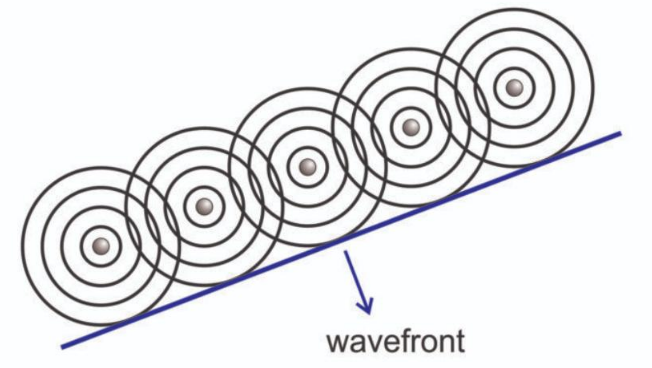
复习波前（wavefront）的概念



a. Scattering of X-rays by an atom

b. circular ripples created by a bee in water

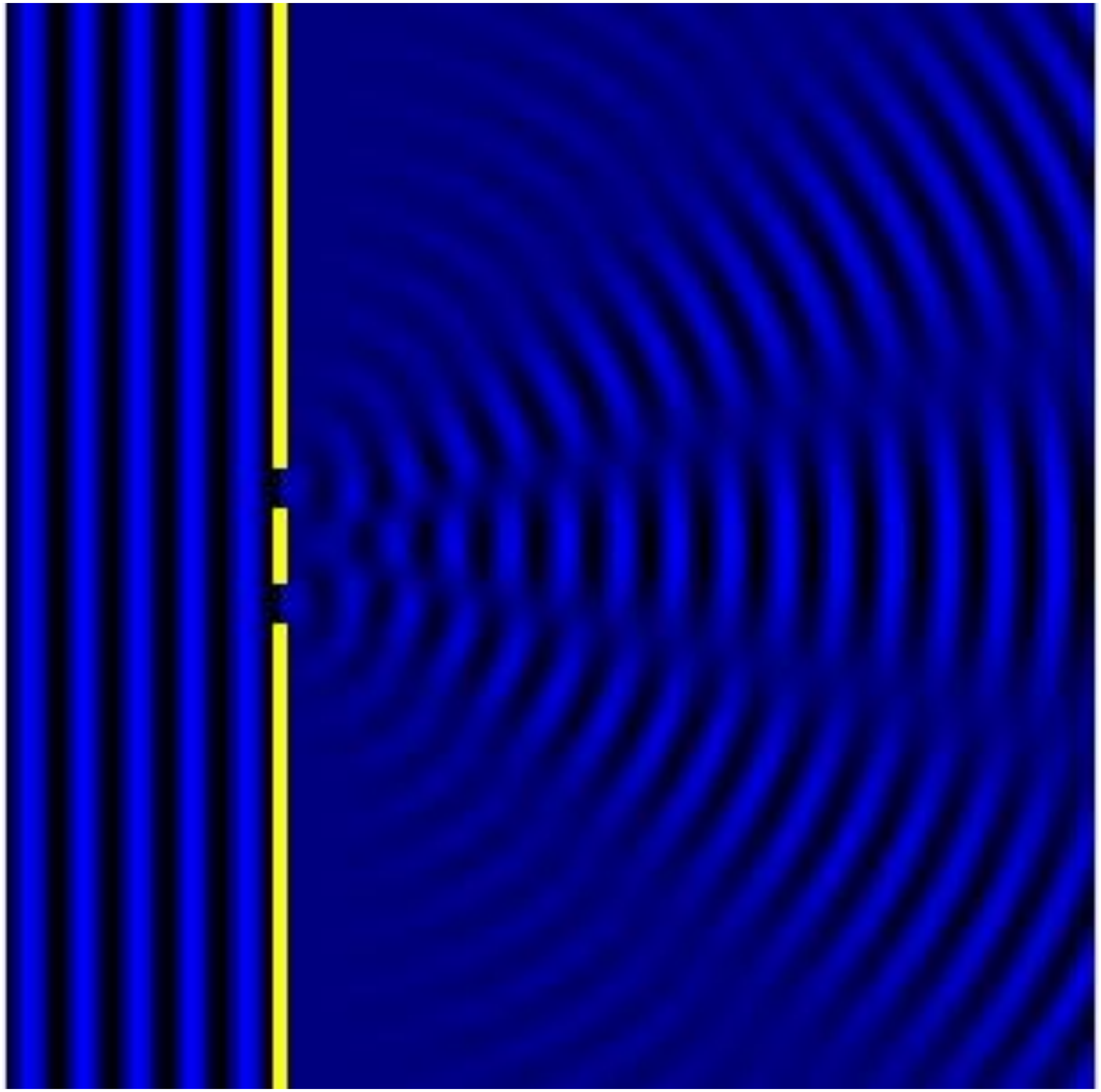


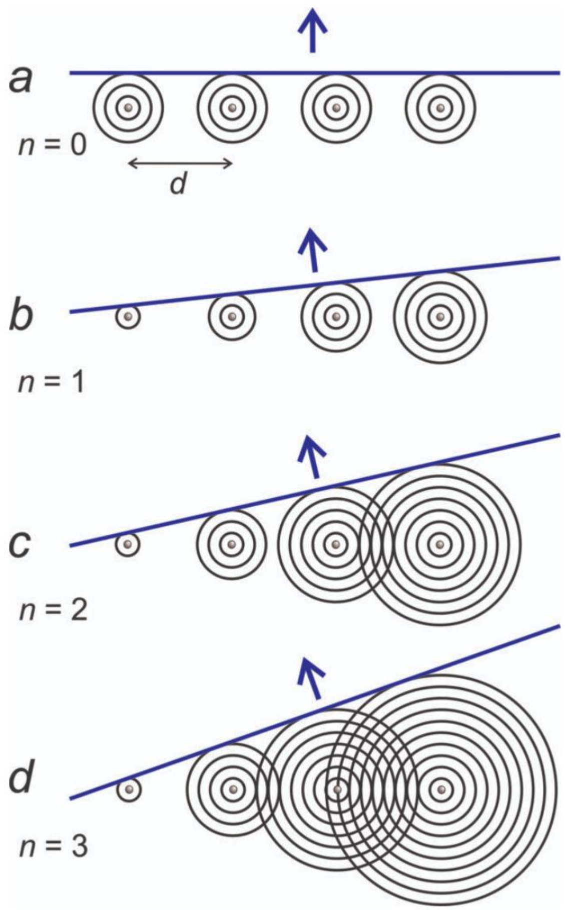


X射线经过原子阵列后发生衍射

波的干涉

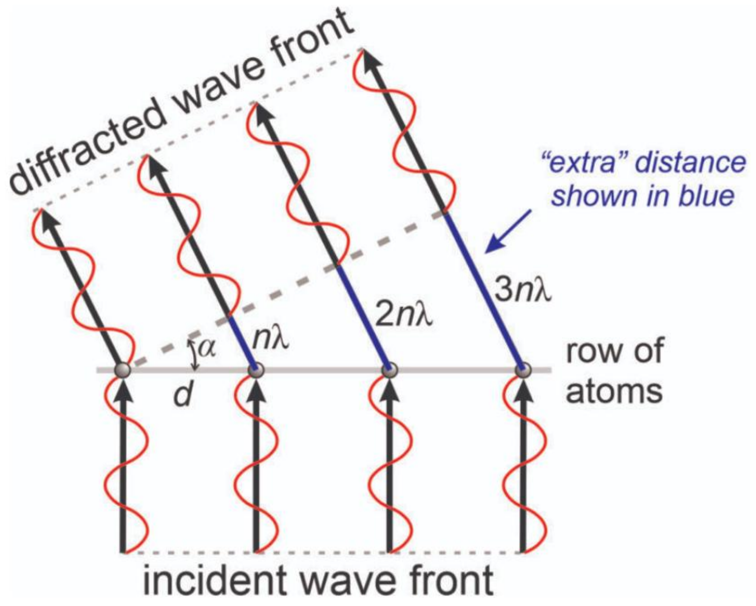
模拟双缝衍射





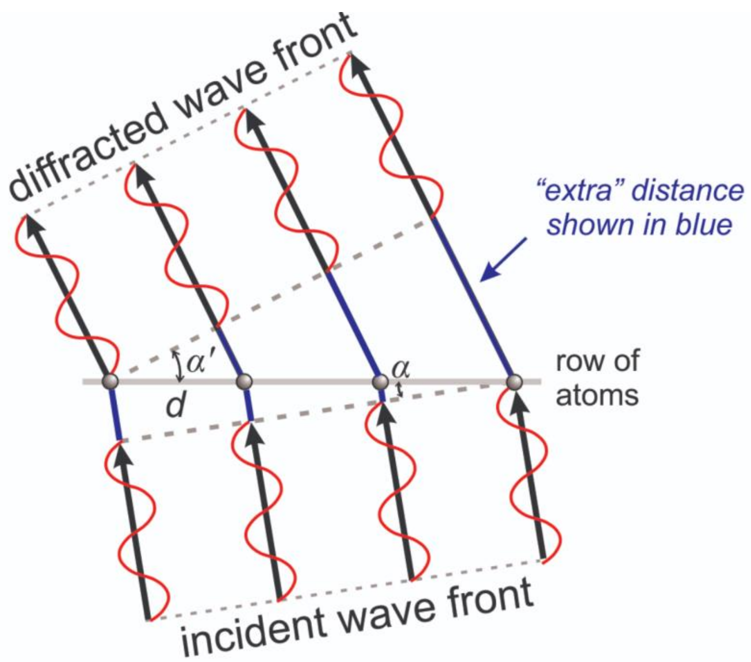
12.12 Higher order diffraction

多级衍射



$$n\lambda = d\sin\alpha$$

垂直入射波



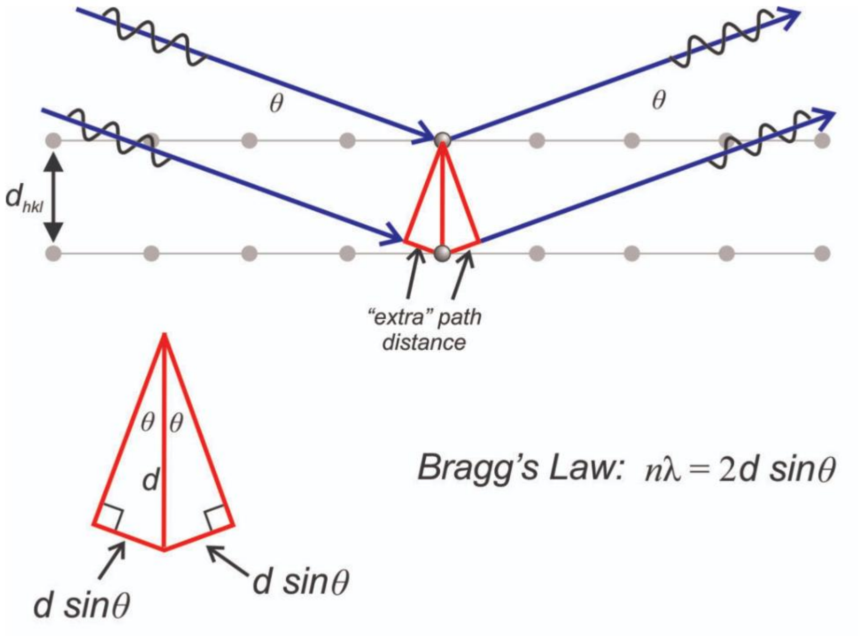
$$n\lambda = d(\sin\alpha' - \sin\alpha)$$

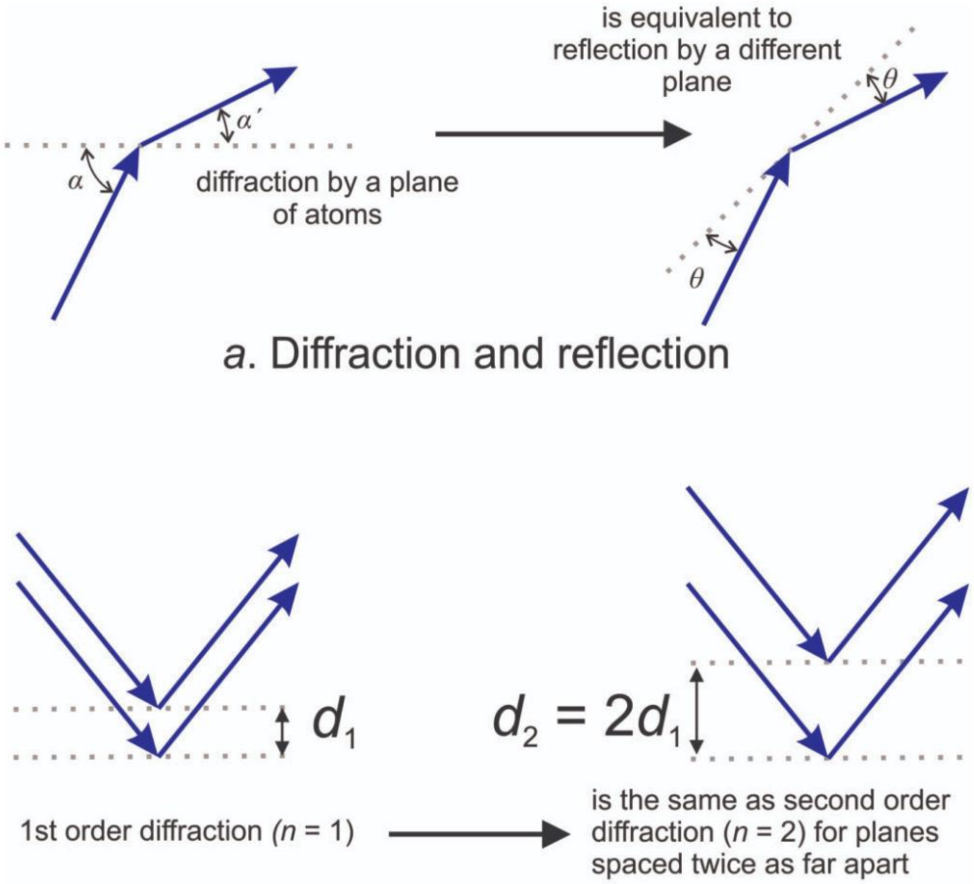
一般情况

Laue（劳埃）方程

布拉格的等价处理

反射角与入射角、衍射角的关系：$\theta = (\alpha - \alpha')/2$





a. Diffraction and reflection

b. First-order and second-order diffraction

衍射等价于“反射”

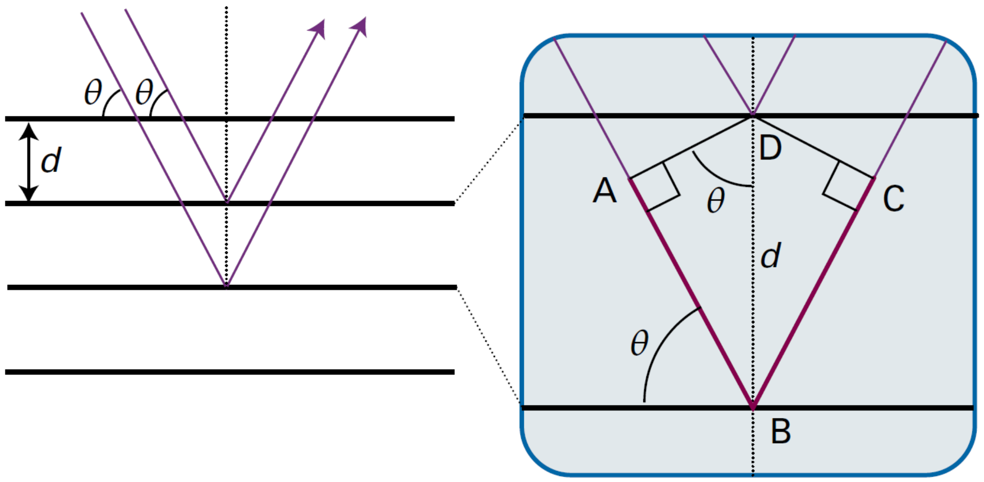
Bragg方程

将晶面看成反射平面

相邻晶面反射光发生
相长干涉的条件

$$n\lambda = 2d\sin\theta$$

$n = 1$是一级反射



2θ是掠射角(glancing angle)

立方晶格

$$\sin\theta = \frac{\lambda}{2d_{hkl}} = (h^2+k^2+l^2)^{1/2}\frac{\lambda}{2a}$$

$d_{hkl} = a/(h^2+k^2+l^2)^{1/2}$

$\sin\theta$ 成特定比例，反映晶格特征

$\{hkl\}$	{100}	{110}	{111}	{200}	{210}
$h^2+k^2+l^2$	1	2	3	4	5
$\{hkl\}$	{211}	{220}	{300}	{221}	{310}
$h^2+k^2+l^2$	6	8	9	9	10

$$h^2+k^2+l^2 \neq 7, 15 \ldots$$

X射线散射

电磁波传播时偏离特定方向（不走直线，各向传播）

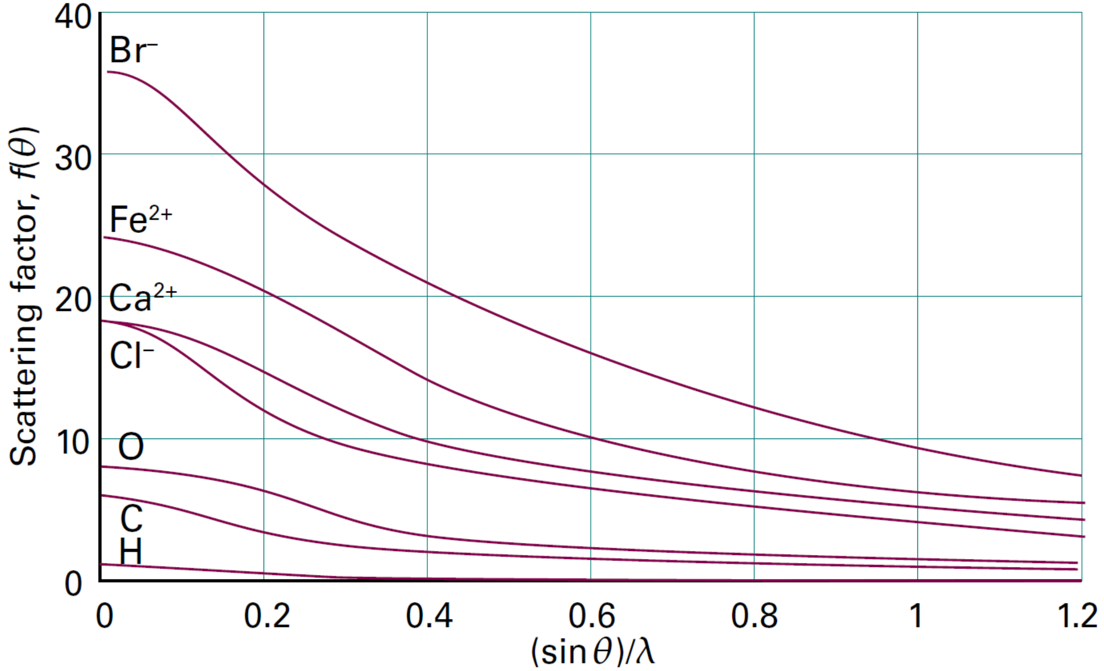
散射因子f：反映散射强度随元素电子数的变化

对于球对称的原子

$$f(\theta)=4\pi\int_0^{\infty}\rho(r)\frac{\sin kr}{kr}r^2\mathrm{d}r \quad k=\frac{4\pi}{\lambda}\sin\theta$$

$$\theta=0,\ f(\theta)=N_e$$

电子越多，散射越强



结构因子F_{hkl}

表征干涉振幅与晶胞结构（原子类型以及位置）的关系

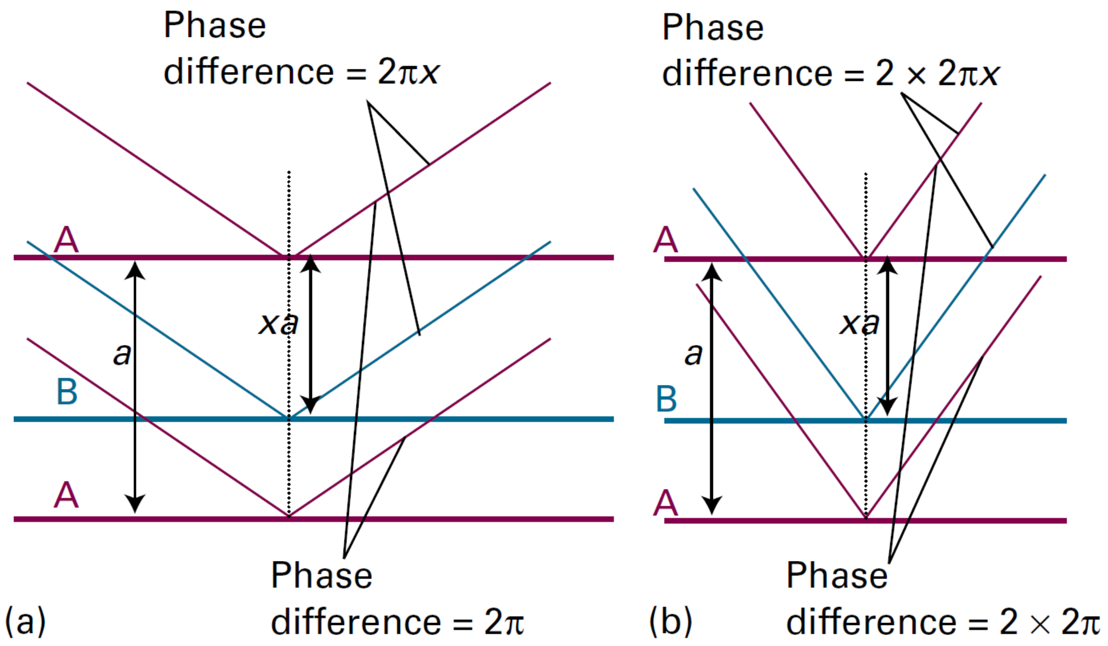
先考虑$(h00)$晶面反射

A原子反射贡献相位差为0

B原子反射贡献相位差为$2h\pi x$

对一般晶面(hkl), B原子贡献相位差为$\phi_{hkl} = 2\pi(hx+ky+lz)$

$$F_{hkl} = f_A + f_B e^{i\phi_{hkl}}$$

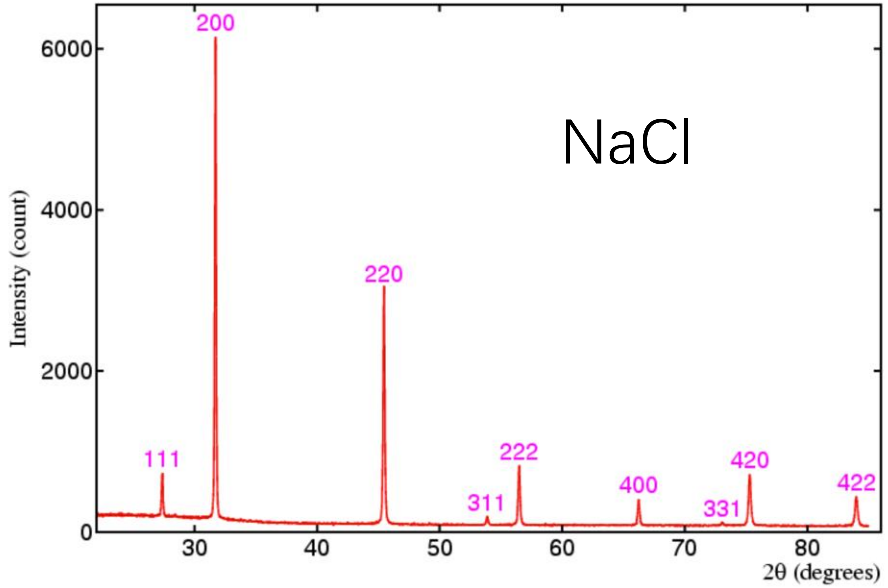


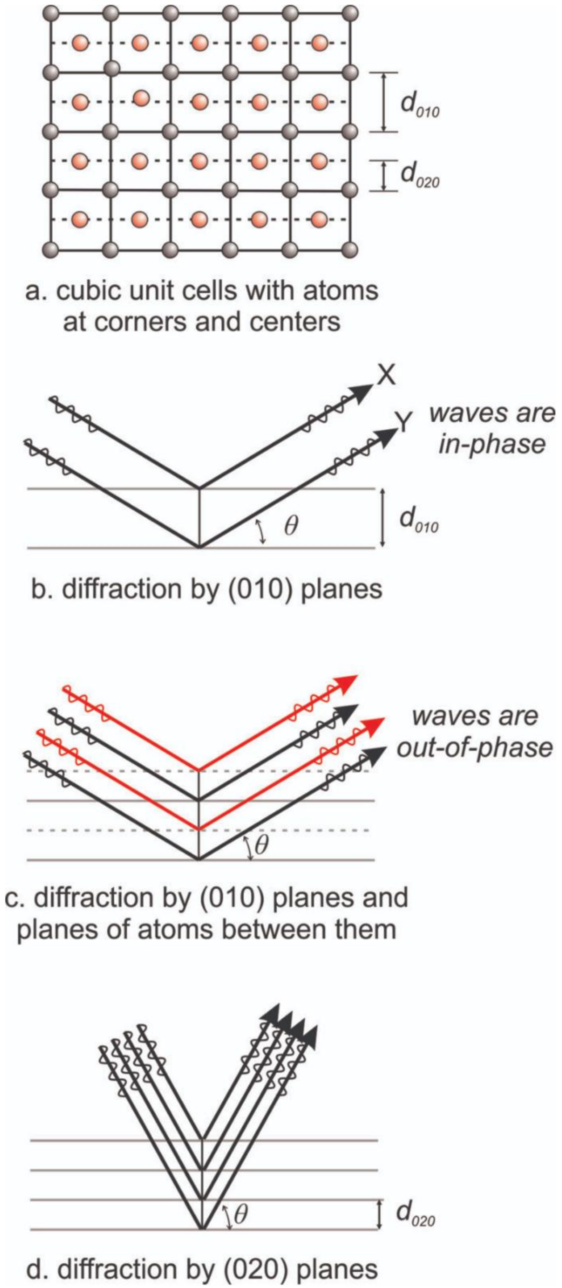
$$F_{hkl} = \sum_j f_j e^{i\phi_{hkl}(j)}$$ 一般情况结构因子表达式

光强正比于振幅模方

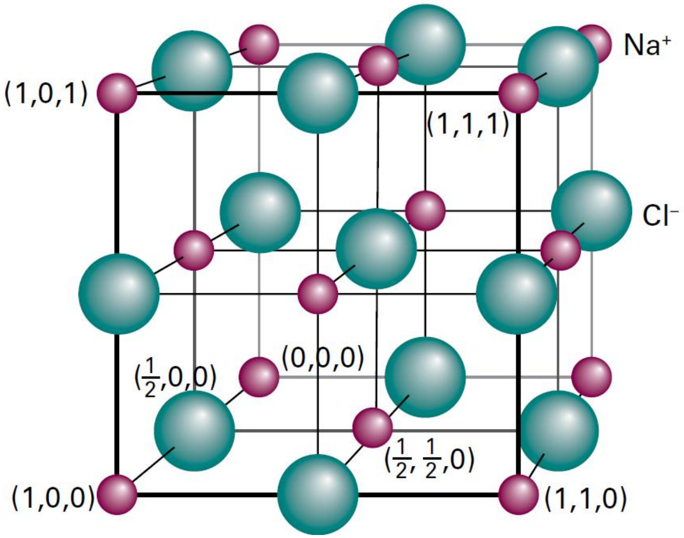
$$I_{hkl} \propto F^{*}_{hkl} F_{hkl}$$

某些晶面反射振幅为0，叫做**系统消光** (systematic absence / extinction)





例：计算NaCl的结构因子

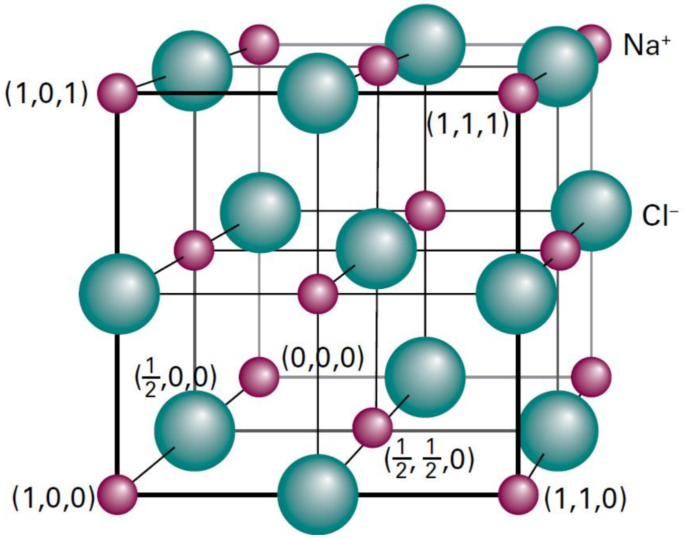


每个离子被多个晶胞共有，需要考虑分数权重

Na^{+}对结构因子贡献：

$$f^{+}[1+(-1)^{h+k}+(-1)^{h+l}+(-1)^{k+l}]$$

Atom	Weight	x	y	z	ϕ_{hkl}	
1	$\frac{1}{8}$	0	0	0	0	1
2	$\frac{1}{8}$	1	0	0	$2\pi h$	
3	$\frac{1}{8}$	0	1	0	$2\pi k$	
4	$\frac{1}{8}$	1	1	0	$2\pi(h+k)$	
5	$\frac{1}{8}$	0	0	1	$2\pi l$	
6	$\frac{1}{8}$	1	0	1	$2\pi(h+l)$	
7	$\frac{1}{8}$	0	1	1	$2\pi(k+l)$	
8	$\frac{1}{8}$	1	1	1	$2\pi(h+k+l)$	
9	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	$2\pi(\frac{1}{2}h+\frac{1}{2}k)$	$(-1)^{h+k}$
10	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	$2\pi(\frac{1}{2}h+\frac{1}{2}l)$	$(-1)^{h+l}$
11	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}$	$2\pi(\frac{1}{2}k+\frac{1}{2}l)$	$(-1)^{k+l}$
12	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$2\pi(h+\frac{1}{2}k+\frac{1}{2}l)$	$(-1)^{k+l}$
13	$\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$	$2\pi(\frac{1}{2}h+k+\frac{1}{2}l)$	$(-1)^{h+l}$
14	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$2\pi(\frac{1}{2}h+\frac{1}{2}k+l)$	$(-1)^{h+k}$



Cl⁻对结构因子贡献：

$$f^{-}[(-1)^{h+k+l}+(-1)^{h}+(-1)^{k}+(-1)^{l}]$$

结构因子：

$$F_{hkl}=f^{+}[1+(-1)^{h+k}+(-1)^{h+l}+(-1)^{k+l}]$$
$$+f^{-}[(-1)^{h+k+l}+(-1)^{h}+(-1)^{k}+(-1)^{l}]$$

Atom	Weight	x	y	z	ϕ_{hkl}
1	$\frac{1}{4}$	$\frac{1}{2}$	0	0	πh
2	$\frac{1}{4}$	0	$\frac{1}{2}$	0	πk
3	$\frac{1}{4}$	1	$\frac{1}{2}$	0	$2\pi(h+\frac{1}{2}k)$
4	$\frac{1}{4}$	$\frac{1}{2}$	1	0	$2\pi(\frac{1}{2}h+k)$
5	$\frac{1}{4}$	0	0	$\frac{1}{2}$	πl
6	$\frac{1}{4}$	1	0	$\frac{1}{2}$	$2\pi(h+\frac{1}{2}l)$
7	$\frac{1}{4}$	0	1	$\frac{1}{2}$	$2\pi(k+\frac{1}{2}l)$
8	$\frac{1}{4}$	1	1	$\frac{1}{2}$	$2\pi(h+k+\frac{1}{2}l)$
9	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$2\pi(\frac{1}{2}h+\frac{1}{2}k+\frac{1}{2}l)$
10	$\frac{1}{4}$	$\frac{1}{2}$	0	1	$2\pi(\frac{1}{2}h+l)$
11	$\frac{1}{4}$	0	$\frac{1}{2}$	1	$2\pi(\frac{1}{2}k+l)$
12	$\frac{1}{4}$	1	$\frac{1}{2}$	1	$2\pi(h+\frac{1}{2}k+l)$
13	$\frac{1}{4}$	$\frac{1}{2}$	1	1	$2\pi(\frac{1}{2}h+k+l)$

容易消光的结构：带心点阵（面心、体心、边心）
螺旋轴、滑移面

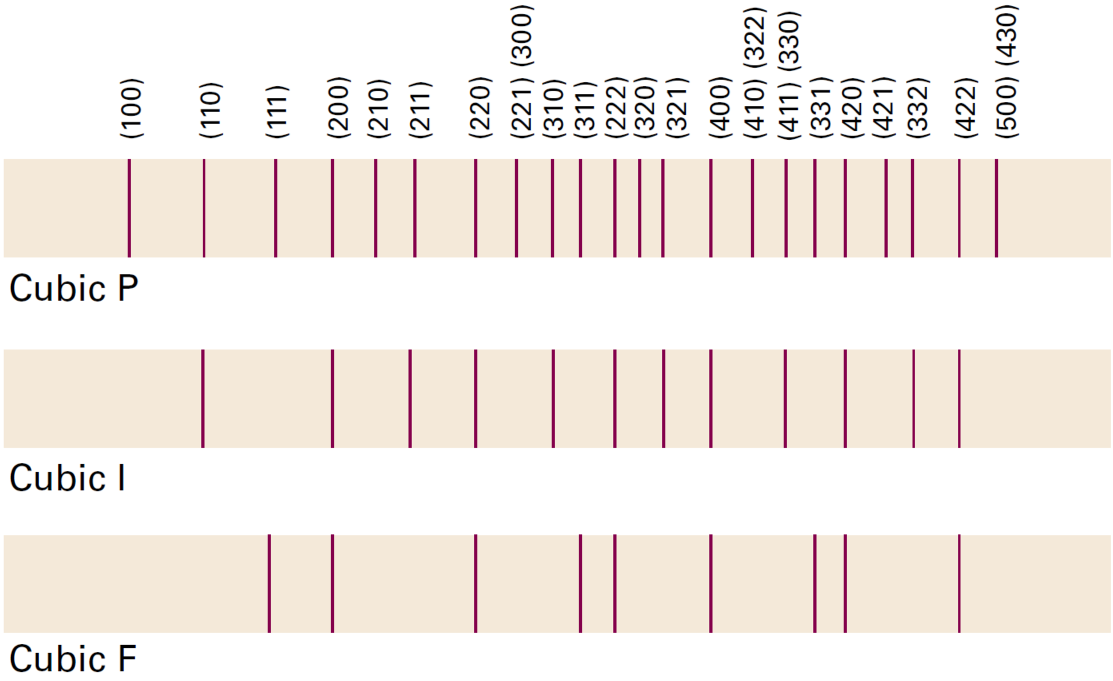
晶面之间有原子

立方面心结构（NaCl）

$$F_{hkl}=f^{+}[1+(-1)^{h+k}+(-1)^{h+l}+(-1)^{k+l}]$$
$$+f^{-}[(-1)^{h+k+l}+(-1)^{h}+(-1)^{k}+(-1)^{l}]$$

系统消光条件：(hkl)

- cP 无
- cI 加和为奇数
- cF 混合奇偶



由结构因子反推电子密度

$$\rho(\boldsymbol{r})=\frac{1}{V}\sum_{hkl}F_{hkl}\mathrm{e}^{-2\pi\mathrm{i}(hx+ky+lz)}$$

注意：位置用原子分数坐标表示，可反推到实空间

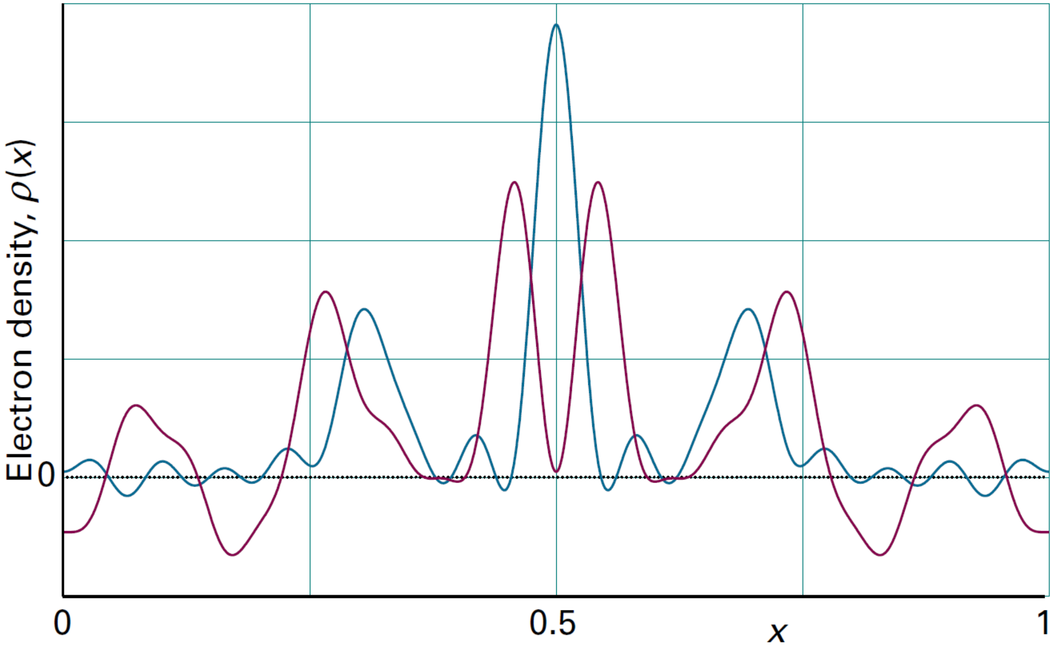
傅里叶变换　V是晶胞体积

例：$\{h00\}$晶面测得结构因子为

h	0	1	2	3	4	5	6	7	8	9
F_h	16	−10	2	−1	7	−10	8	−3	2	−3
h	10	11	12	13	14	15				
F_h	6	−5	3	−2	2	−3				

且有 $F_{-h}=F_h$

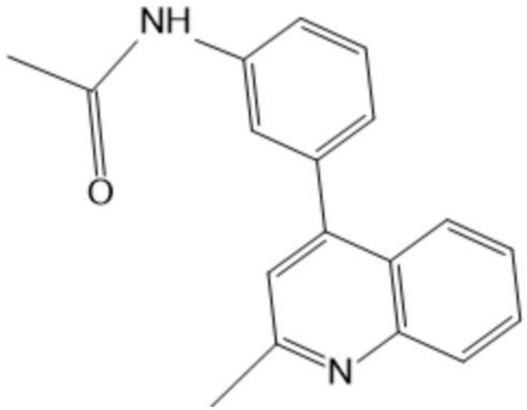
画出x方向的电子密度

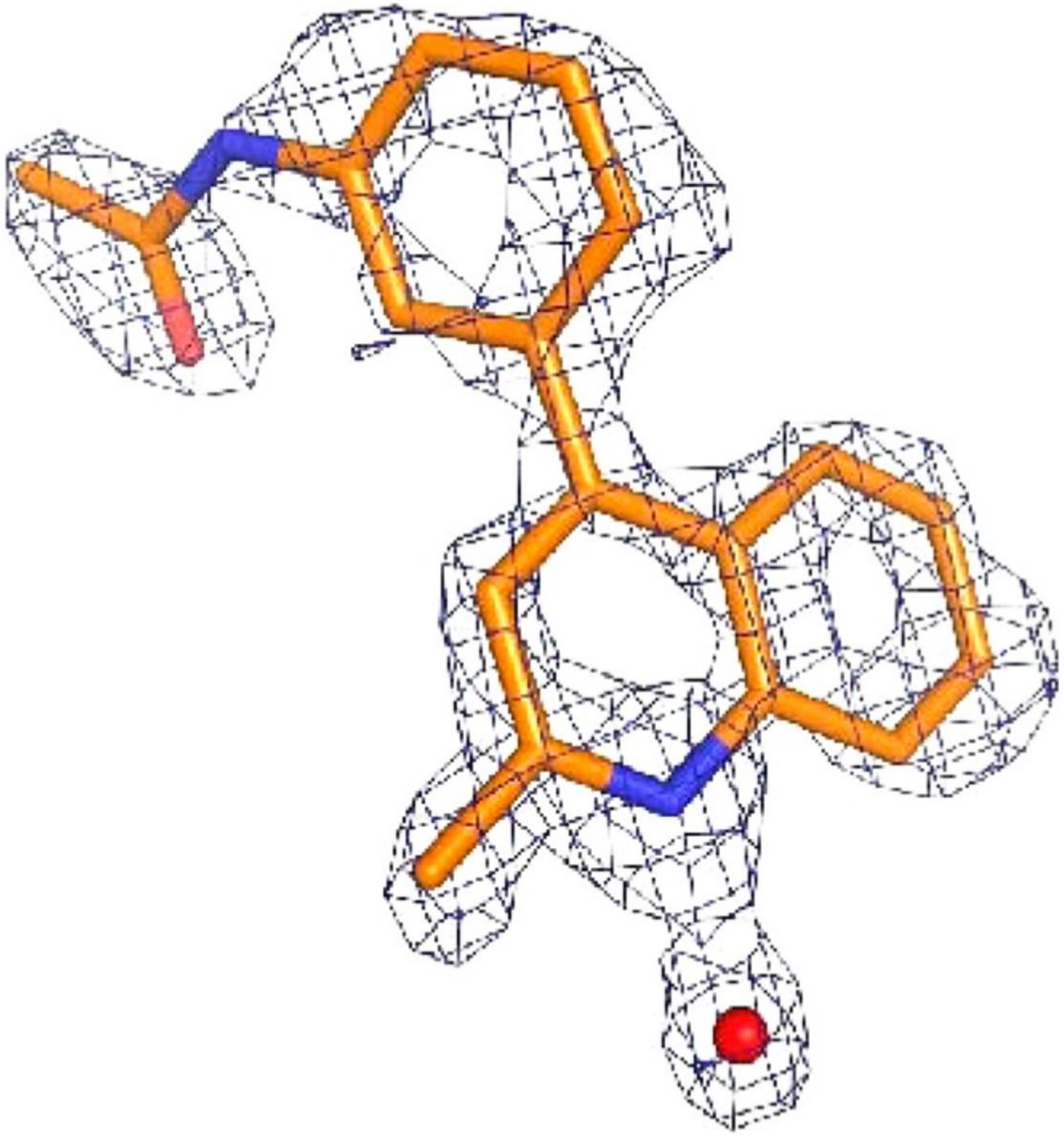


$$V\rho(x)=\sum_{h=-\infty}^{\infty}F_h\mathrm{e}^{-2\pi\mathrm{i}hx}=F_0+\sum_{h=1}^{\infty}(F_h\mathrm{e}^{-2\pi\mathrm{i}hx}+\overbrace{F_{-h}}^{=F_h}\mathrm{e}^{2\pi\mathrm{i}hx})\overset{\mathrm{e}^{\mathrm{i}x}+\mathrm{e}^{-\mathrm{i}x}=2\cos x}{=}F_0+2\sum_{h=1}^{\infty}F_h\cos2\pi hx$$

展开项不够(只有15项)，密度不够精确（绿线）

根据X射线衍射数据
重构出喹啉衍生物的
电子密度等值面图



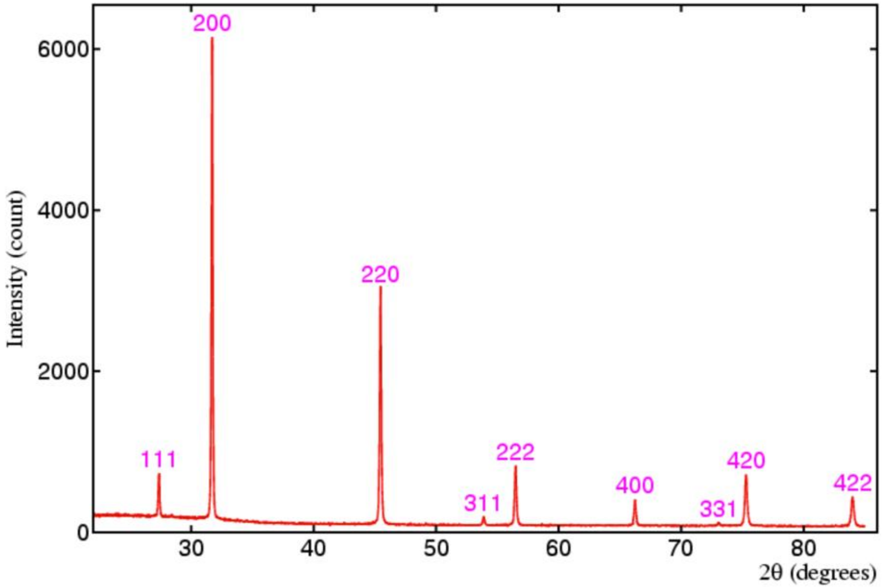


Lewis R. Vidler et al. J. Med. Chem. 56, 8073–8088, 2013.

XRD解析晶体结构流程

以立方晶系为例

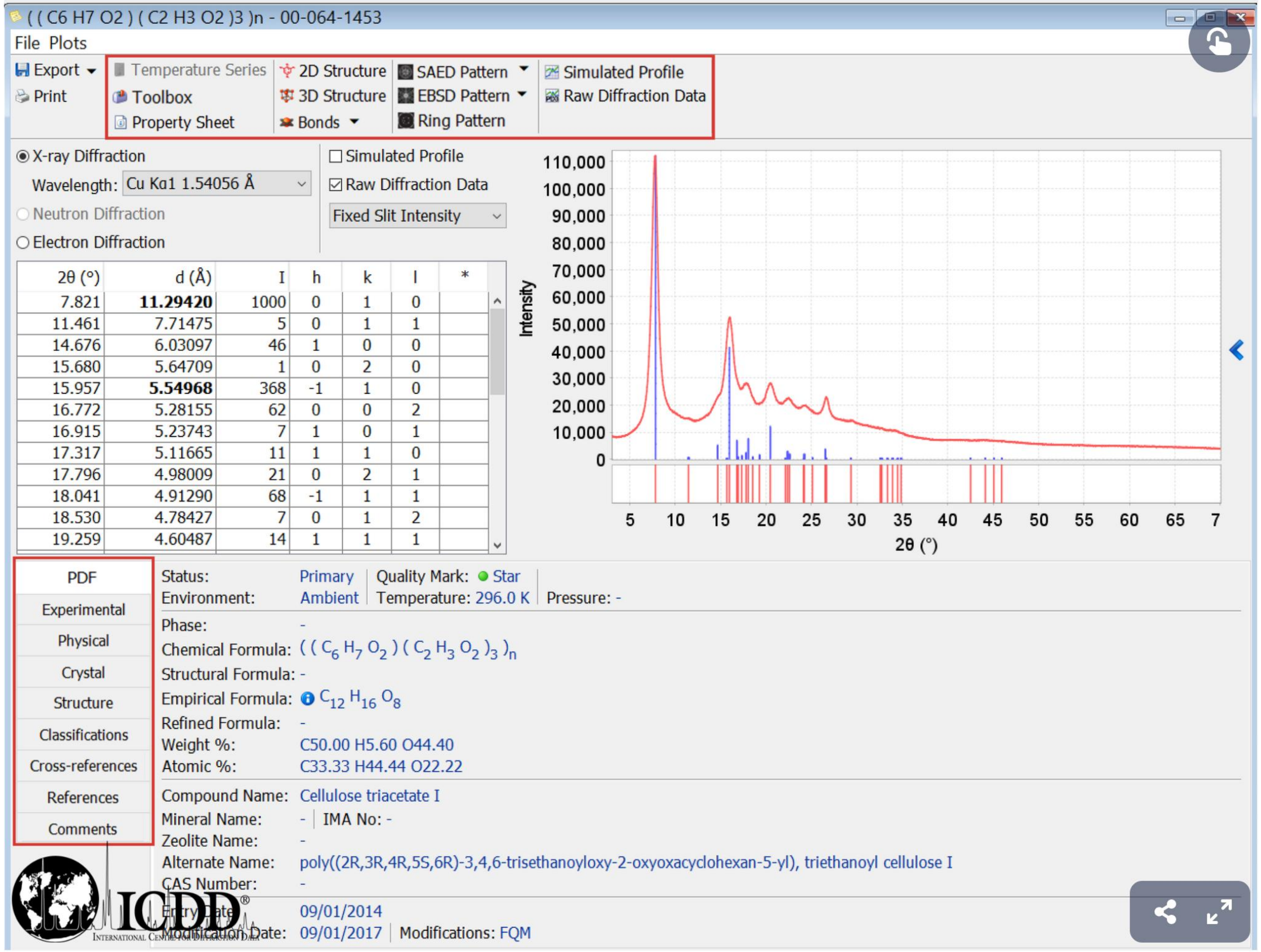
- 记录衍射方向、强度，读出衍射峰对应的2θ
- 由$\sin^2\theta$算出$h^2+k^2+l^2$
- 根据消光判断点阵
- 由X射线波长与$\sin\theta$推出晶面间距d
- 由 d 与 $h^2+k^2+l^2$算出晶胞参数
- 由晶胞体积与电子密度推出结构基元



X-ray Powder Diffraction File (PDF)

International Center for Diffraction Data

https://www.icdd.com/



PDF | Experimental | Physical | Crystal | **Structure** | Classifications | Cross-references | References | Comments

AC Space Group: P1121 (4)

AC Unit Cell

a: 6.060(5) Å α: 90.0°

b: 11.3483(31) Å β: 90.0°

c: 10.563(11) Å γ: 95.60(8)°

Space Group Symmetry Operators (2)

Seq	Operator
1	x,y,z
2	-x,-y,z+1/2

Atomic Coordinates (36) ADP Type: B U Origin: -

Atom	Num	Wyckoff	Symmetry	x	y	z	SOF	IDP	AET
C	1			0.023	0.0323	0.3969	1.0		
C	2			0.2064	0.0865	0.309	1.0		
C	3			0.4832	0.2517	0.348	1.0		
C	4			0.5352	0.3713	0.4058	1.0		
C	5			0.1253	0.0924	0.1715	1.0		

Anisotropic Displacement Parameters (0)

Crystal (Symmetry Allowed): Non-centrosymmetric - Enantiomorphic, Optical Activity, Pyro / Piezo (p), Piezo (2nd Harm.)

PDF | Experimental | Physical | **Crystal** | Structure | Classifications | Cross-references | References | Comments

ICDD Calculated Parameters

Space Group: P21 (4)

Molecular Weight: 288.25 g/mol

Crystal Data

a: 11.348 Å	α: 90.00°	Volume: 722.96 Å³	c/a: 0.534
b: 10.563 Å	β: 95.60°	Z: 2.00	a/b: 1.074
c: 6.060 Å	γ: 90.00°		c/b: 0.574

Reduced Cell

a: 6.060 Å	α: 90.00°	Volume: 722.96 Å³
b: 10.563 Å	β: 95.60°	
c: 11.348 Å	γ: 90.00°	

中子衍射

中子波长与X射线相当

估算373K热平衡时中子波长

de Broglie关系： $\lambda = \dfrac{h}{p}$

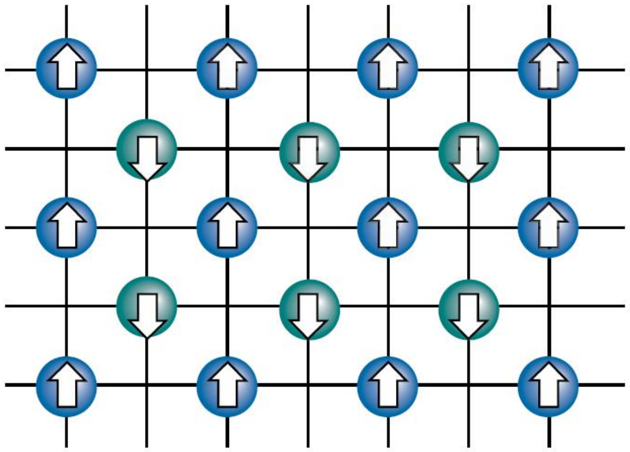
假设中子在x方向运动，

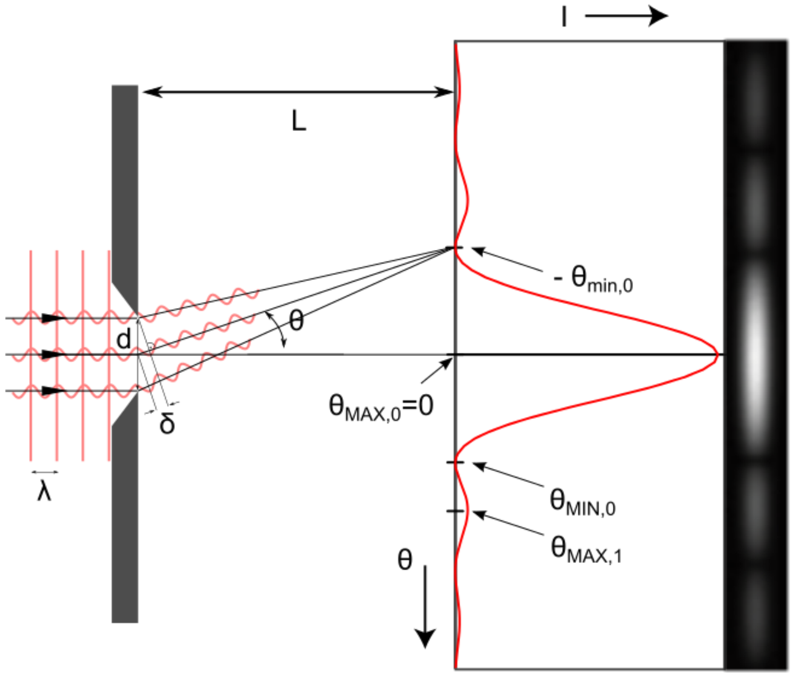
$$E_k = \frac{1}{2}kT = \frac{p^2}{2m}$$

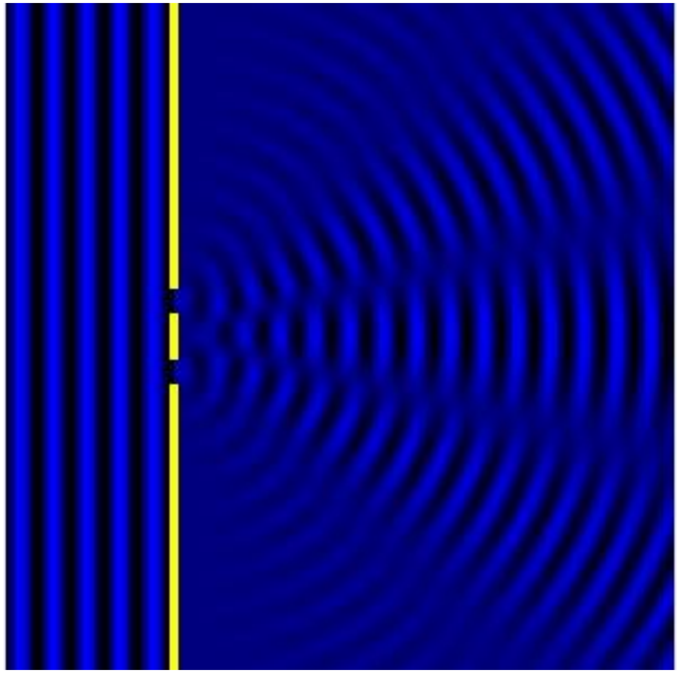
$$\lambda = \frac{h}{\sqrt{mkT}} = 226\text{pm}$$

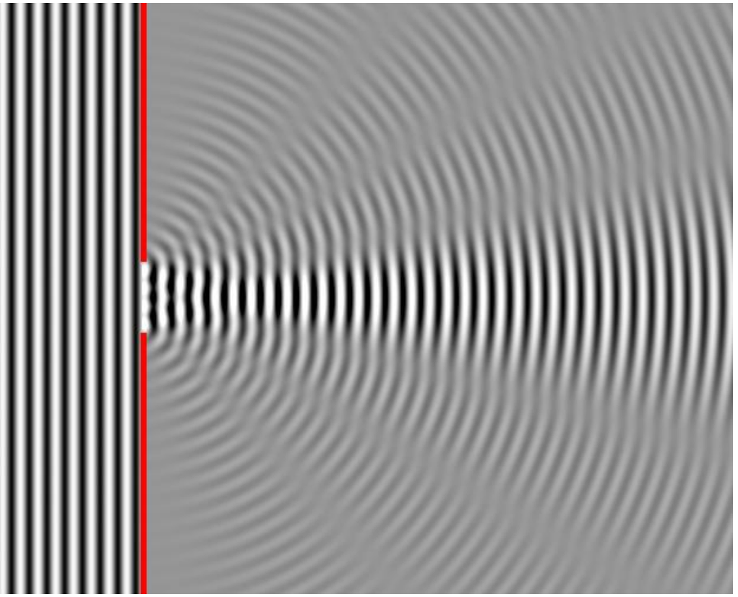
中子衍射特点：

- 与原子电子数无关
- 利用中子与原子核的相互作用，受临近原子影响，可区分相邻元素
- 中子有磁矩，适合研究磁性材料









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